Market Insights

Real-Time Market Intelligence Pipeline with NLP and RAG

Learning Resources and Contributor Onboarding

Doctum Consilium

Generated 2026-05-08

Abstract

This document provides a curated list of courses (Coursera, Pluralsight, A Cloud Guru), training repositories, and reference books to help contributors ramp up on the technologies used in **Market Insights**. Resources are selected for quality, relevance, and practical applicability.

Contents

Prerequisites

Before starting the courses below, ensure you are comfortable with: Python intermediate, SQL basics, basic NLP concepts, Docker fundamentals.

Coursera Courses

Course	Link	Why relevant
Natural Language Processing Specialization	link	Tokenisation, embeddings, seq2seq models used in the NLP pipeline.
Data Engineering with Google Cloud	link	Pipeline patterns, batch vs streaming — Airflow parallels.
IBM Data Engineering Professional Certificate	link	SQL, ETL, Airflow, Kafka fundamentals.

Table 1: Coursera

Pluralsight Courses

Course	Link	Why relevant
Apache Airflow: The Hands-On Guide	link	DAG authoring, scheduling, XCom, sensors.
Applied Natural Language Processing in Python	link	Text preprocessing, TF-IDF, embeddings.

Table 2: Pluralsight

A Cloud Guru Courses

Course	Link	Why relevant
AWS Data Analytics Specialty	link	Data pipeline patterns applicable to this architecture.

Table 3: A Cloud Guru

GitHub Training Repositories

- **apache/airflow** — Source, examples, and DAG best practices.
- **facebookresearch/faiss** — Vector index internals; understand IVF, HNSW index types.
- **UKPLab/sentence-transformers** — Embedding models used for semantic search.
- **langchain-ai/langchain** — RAG pipeline patterns and retriever abstractions.

Reference Books

- *Designing Data-Intensive Applications* — Martin Kleppmann. Foundational for pipeline reliability and data consistency.
- *Natural Language Processing with Transformers* — Lewis Tunstall et al.. Transformer models, fine-tuning, embeddings.

Suggested Learning Path

1. Complete prerequisites (1–2 weeks).
2. Work through the Coursera specialisation most relevant to the repo's domain (4–8 weeks).
3. Clone the repository, read `README.md` and `INFRASTRUCTURE.md`.
4. Run the service locally following the **Local Run** section of `INFRASTRUCTURE.md`.
5. Study the Pluralsight courses for the specific tools used (2–4 weeks, parallel).
6. Contribute a small fix or documentation improvement as a first PR.
7. Study the GitHub training repositories for advanced patterns.

Free Resources

- **Python Official Tutorial** — free, authoritative.
- **Kubernetes Interactive Tutorials** — browser-based k8s labs.
- **FastAPI Tutorial** — official, comprehensive.
- **PyTorch Tutorials** — official, covers all levels.
- **DeepLearning.AI Short Courses** — free, recent AI topics.